

EXPERIMENT 2

AIM

To add the constraints like **PRIMARY KEY**, **FOREIGN KEY**, **UNIQUE KEY**, and **CHECK** using Data Definition Language (DDL) commands.

Description

PRIMARY KEY

- The **PRIMARY KEY** constraint uniquely identifies each record in a table.
- Primary key values must be **UNIQUE** and **NOT NULL**.
- A table can have only **one PRIMARY KEY**.

FOREIGN KEY

- A **FOREIGN KEY** is used to establish a relationship between two tables.
- It refers to the **PRIMARY KEY** of another table.
- The table containing the foreign key is called the **child table**, and the referenced table is called the **parent table**.

UNIQUE Constraint

- The **UNIQUE** constraint ensures that all values in a column are different.
- A table can have multiple **UNIQUE** constraints.
- A **PRIMARY KEY** automatically has a **UNIQUE** constraint.

CHECK Constraint

- The **CHECK** constraint restricts the values that can be stored in a column.
- It ensures that only valid data is inserted into the table.

Syntax

PRIMARY KEY

```
ALTER TABLE table_name  
ADD PRIMARY KEY(column_name);
```

FOREIGN KEY

```
ALTER TABLE table_name  
ADD CONSTRAINT constraint_name  
FOREIGN KEY(column_name)  
REFERENCES parent_table(column_name);
```

UNIQUE

```
ALTER TABLE table_name  
ADD CONSTRAINT constraint_name  
UNIQUE(column_name);
```

CHECK

```
ALTER TABLE table_name  
ADD CONSTRAINT constraint_name  
CHECK(condition);
```

Questions

1) Alter the table PATIENT with the following structure.

S.No Column Name Constraint

1	PatientID	PRIMARY KEY
---	-----------	-------------

2) Alter the table DOCTOR with the following structure.

S.No Column Name Constraint

1	DoctorID	PRIMARY KEY
2	Gender	CHECK ('M' or 'F')

3) Alter the table DEPARTMENT_DETAILS with the following structure.

S.No Column Name Constraint

1 DeptID PRIMARY KEY

4) Alter the table APPOINTMENT_DETAILS with the following structure.

S.No Column Name Constraint

1 AppointmentID PRIMARY KEY

2 RoomNo CHECK (Room Number between 1 and 50)

5) Alter the table PATIENT with the following structure.

S.No Column Name Constraint

1 MobileNo UNIQUE KEY

6) After the PATIENT, DOCTOR, and APPOINTMENT_DETAILS tables are successfully created, add FOREIGN KEY constraints to the APPOINTMENT_DETAILS table.

S.No	Child Table	Foreign Key	Parent Table	Primary Key
------	-------------	-------------	--------------	-------------

1	APPOINTMENT_DETAILS	PatientID	PATIENT	PatientID
---	---------------------	-----------	---------	-----------

2	APPOINTMENT_DETAILS	DoctorID	DOCTOR	DoctorID
---	---------------------	----------	--------	----------

Question 1

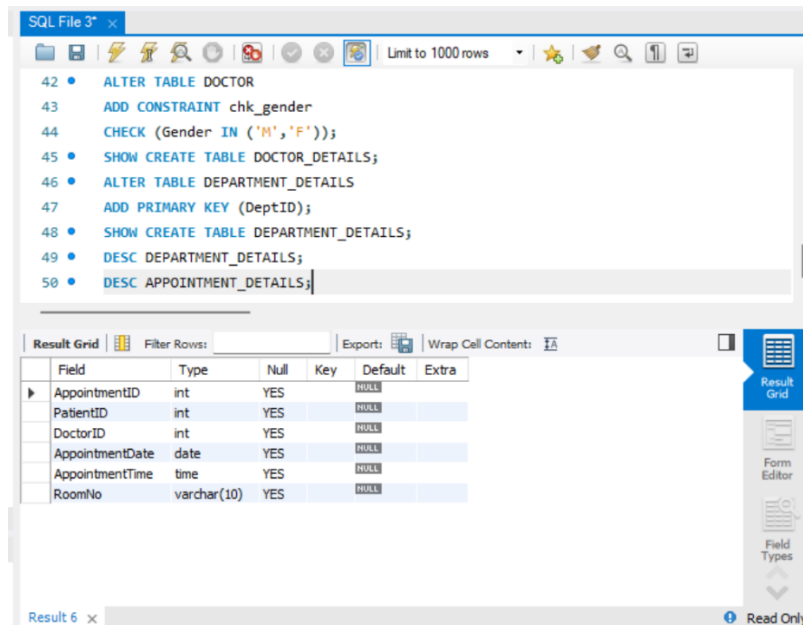
Alter the table PATIENT by adding PatientID as the PRIMARY KEY.

```
ALTER TABLE PATIENT
```

```
ADD PRIMARY KEY (PatientID);
```

Verification

```
DESC PATIENT;
```



Question 2

Alter the table DOCTOR by adding DoctorID as the PRIMARY KEY and Gender as a CHECK constraint ('M' or 'F').

Step 1: Add Primary Key

```
ALTER TABLE DOCTOR
```

```
ADD PRIMARY KEY (DoctorID);
```

Step 2: Add CHECK Constraint

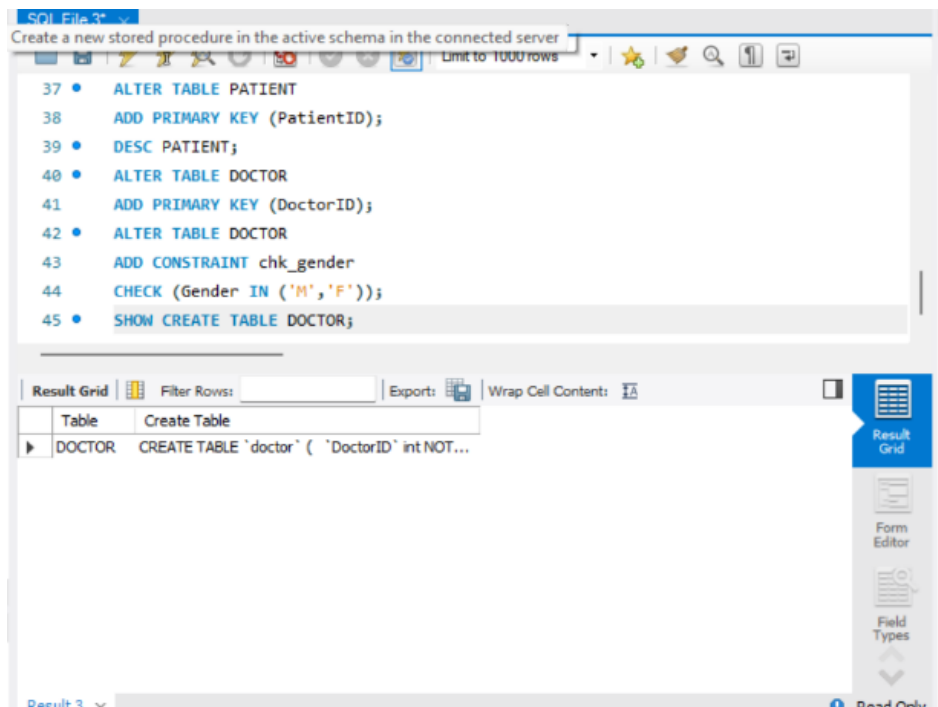
```
ALTER TABLE DOCTOR
```

```
ADD CONSTRAINT chk_gender
```

```
CHECK (Gender IN ('M','F'));
```

Verification

```
SHOW CREATE TABLE DOCTOR;
```



Question 3

Alter the table DEPARTMENT_DETAILS by adding DeptID as the PRIMARY KEY.

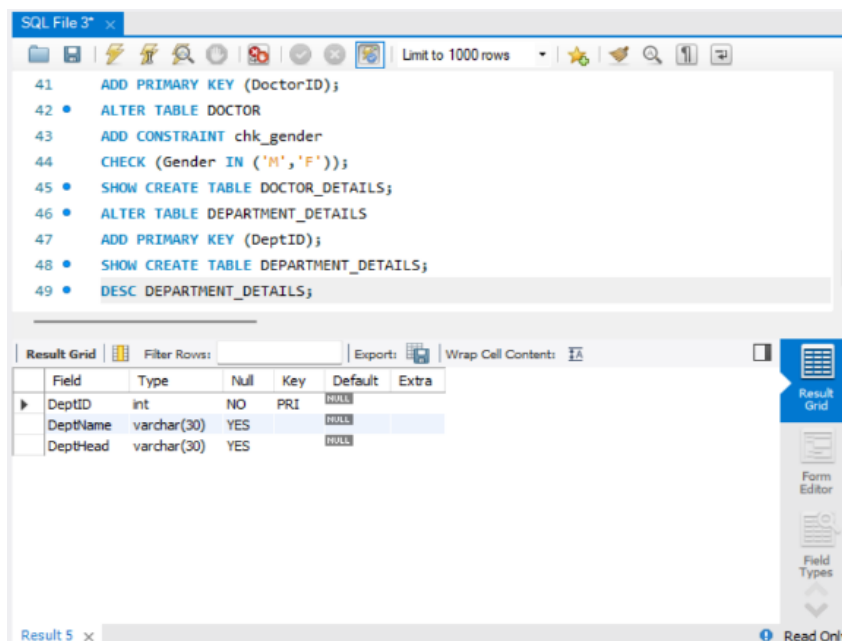
ALTER TABLE DEPARTMENT_DETAILS

ADD PRIMARY KEY (DeptID);

Verification

DESC DEPARTMENT_DETAILS;

SHOW CREATE TABLE DEPARTMENT_DETAILS;



Question 4

Alter the table **APPOINTMENT_DETAILS** by adding **AppointmentID** as the **PRIMARY KEY** and a **CHECK** constraint for **RoomNo** (between 1 and 50).

Step 1: Add Primary Key

```
ALTER TABLE APPOINTMENT_DETAILS
```

```
ADD PRIMARY KEY (AppointmentID);
```

Step 2: Add CHECK Constraint

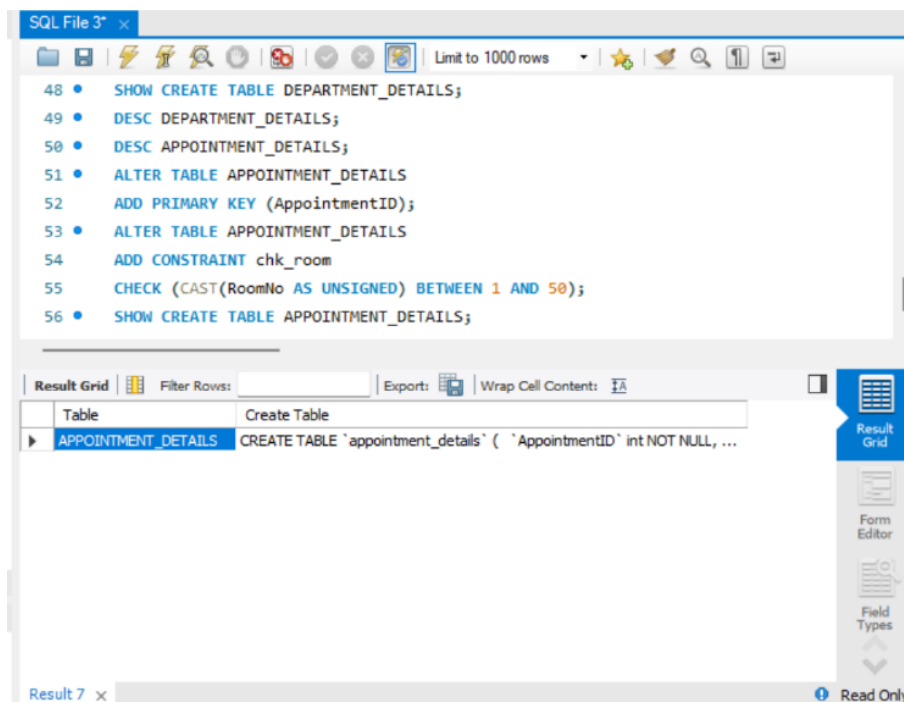
```
ALTER TABLE APPOINTMENT_DETAILS
```

```
ADD CONSTRAINT chk_room
```

```
CHECK (CAST(RoomNo AS UNSIGNED) BETWEEN 1 AND 50);
```

Verification

```
SHOW CREATE TABLE APPOINTMENT_DETAILS;
```



Question 5

Alter the table **PATIENT** by adding a **UNIQUE** constraint to **MobileNo**.

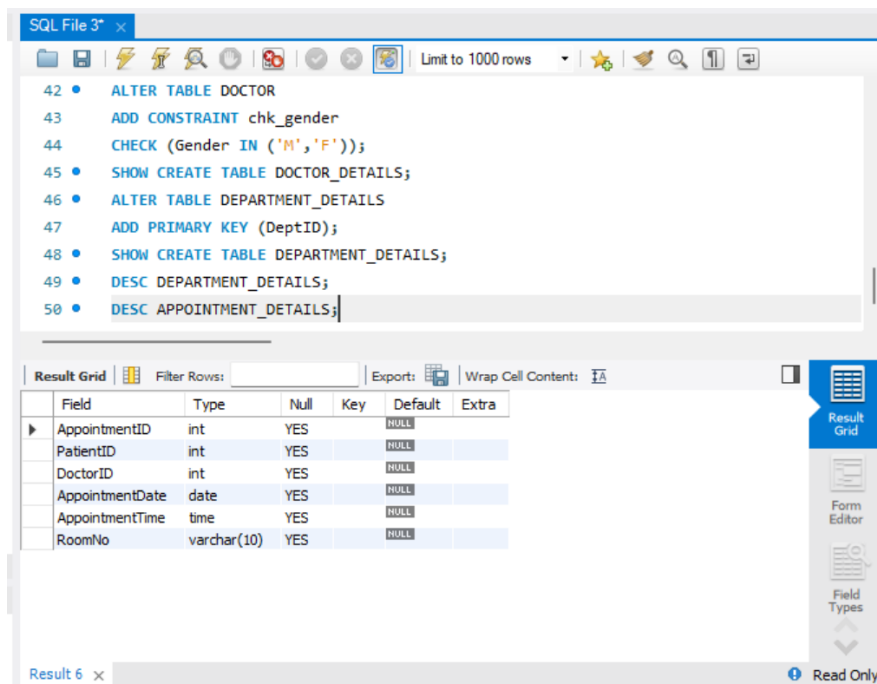
```
ALTER TABLE PATIENT
```

```
ADD CONSTRAINT uq_mobile
```

```
UNIQUE (MobileNo);
```

Verification

SHOW CREATE TABLE PATIENT;



Question 6

Add FOREIGN KEY constraints to the APPOINTMENT_DETAILS table.

Step 1: Add Foreign Key for PatientID

```
ALTER TABLE APPOINTMENT_DETAILS
```

```
ADD CONSTRAINT fk_patient
```

```
FOREIGN KEY (PatientID)
```

```
REFERENCES PATIENT(PatientID);
```

Step 2: Add Foreign Key for DoctorID

```
ALTER TABLE APPOINTMENT_DETAILS
```

```
ADD CONSTRAINT fk_doctor
```

```
FOREIGN KEY (DoctorID)
```

```
REFERENCES DOCTOR(DoctorID);
```

Verification

SHOW CREATE TABLE APPOINTMENT_DETAILS;

The screenshot shows the MySQL Workbench interface. The SQL editor window contains the following SQL commands:

```
69
70 • SHOW CREATE TABLE DOCTOR;
71 • ALTER TABLE APPOINTMENT_DETAILS
72   ADD CONSTRAINT fk_patient
73   FOREIGN KEY (PatientID)
74   REFERENCES PATIENT(PatientID);
75 • ALTER TABLE APPOINTMENT_DETAILS
76   ADD CONSTRAINT fk_doctor
77   FOREIGN KEY (DoctorID)
78   REFERENCES DOCTOR(DoctorID);
79 • SHOW CREATE TABLE APPOINTMENT_DETAILS;
80 • SELECT * FROM PATIENT;
81 • INSERT INTO DOCTOR
82   VALUES
83   (201,'Dr. Kumar','H','Cardiology',9876501234,10);
84 • INSERT INTO APPOINTMENT_DETAILS
85   VALUES
86   (1,101,201,'2026-07-29','10:30:00','12');
87 • INSERT INTO APPOINTMENT_DETAILS
88   VALUES
89   (2,999,201,'2026-07-30','11:00:00','15');
```

The output window at the bottom shows the execution results:

#	Time	Action	Message
83	11:32:43	INSERT INTO APPOINTMENT_DETAILS VALUES (1,101,201,2026-07-29,'10:30...	1 row(s) affected
84	11:32:51	INSERT INTO APPOINTMENT_DETAILS VALUES (2,999,201,2026-07-30,'11:00...	Error Code: 1452. Cannot add or update a child row: a foreign key constraint

Result

The **PRIMARY KEY**, **FOREIGN KEY**, **UNIQUE**, and **CHECK** constraints were successfully added to the **PATIENT**, **DOCTOR**, **DEPARTMENT_DETAILS**, and **APPOINTMENT_DETAILS** tables using DDL commands. The constraints were verified successfully in MySQL Workbench.